Yezi Chen

Master's Student | Climate Economics, Integrated Assessment Modeling, and Industrial Decarbonization

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RESEARCH PROFILE

Master's student specializing in GCAM-based climate policy evaluation, industrial decarbonization, technology-support policy, and carbon-price feedbacks. Experienced in independent scenario design, model configuration, model execution, database queries, quantitative analysis, visualization, and research writing. Expected graduation: June 2027. Prospective PhD applicant for Fall 2027.

EDUCATION

Guangxi University, School of Business Administration — **2024 - 2027 (expected)**
Academic Master's Program in Business Administration

Research focus: GCAM-based climate policy, industrial decarbonization, and carbon-price feedbacks

Hefei University of Technology, School of Economics — **2019 - 2023**
Bachelor's Degree in Financial Engineering

Selected coursework: Financial Simulation and Modeling, Financial Data Mining, and Machine Learning

CURRENT INDEPENDENT RESEARCH

Low-Carbon Technology Support, Industrial Emissions, and Carbon-Price Feedbacks — **Sep 2025 - Present**
Independent research project

- Use the Global Change Analysis Model (GCAM) to evaluate how the timing and intensity of low-carbon technology support affect industrial emissions, endogenous carbon prices, cumulative emissions, and energy substitution.
- Construct scenarios with support starting in 2025, 2035, and 2045; multiple levels of technology-cost reduction; and unconstrained, common CO_2-constrained, and fixed-carbon-price comparisons.
- Analyze industrial sectors, regional mitigation burdens, energy-input substitution, and the interaction between technology support and carbon-pricing pressure.
- Preliminary result: stronger technology support can reduce carbon-price pressure without monotonically deepening cumulative industrial emissions reductions; stable mitigation also requires the phase-out of high-carbon inputs.
- Independently conduct research design, GCAM learning and configuration, scenario construction, model runs, database queries, output processing, visualization, interpretation, and manuscript preparation.

RESEARCH EXPERIENCE

Cost-Benefit Analysis of Emissions Reduction in China's Carbon Market — **Sep 2024 - Present**
Research participant

- Reproduced and studied DICE, RICE, RICE-N, PyRICE, RICEx50, and RICE2020 using GAMS.
- Contributed to model documentation, literature review, scenario-setting discussions, and research on carbon-price pathways and mitigation costs.

Frontiers in Climate Change Economics — **Jul 2025 - Present**
Research support, Prof. Zhifu Mi's team

- Support literature collection and synthesis on climate economics, low-carbon transition, and related policy questions.

Public Participation in Carbon-Emission Trading Investment — **Undergraduate**
Undergraduate research project leader

- Led questionnaire design, data collection, sample organization, and statistical analysis for a survey-based project on public participation in carbon-emission trading investment.

ACADEMIC TRAINING

ARCH Summer School 2026: Air-Climate-Health, Peking University **Aug 22 - 30, 2026**

Incoming participant

Planned training in atmospheric science, climate change, environmental health, box modeling, particulate-matter source apportionment, and AI applications.

MEIC & CNCAP Summer School 2025 **Aug 14 - 16, 2025**

Participant

Training in multiscale emissions inventories, carbon-neutrality and clean-air co-benefit assessment, and science-based decision support.

QUANTITATIVE PROJECTS

- **Commercial-bank efficiency:** applied DEA-BCC, the Malmquist productivity index, factor analysis, and Tobit models to study efficiency and influencing factors during the COVID-19 period.
- **Financial econometrics:** completed hedging regressions, optimal hedge-ratio estimation, variable construction, model estimation, and significance interpretation using EViews.
- **Python financial data analysis:** processed options data and evaluated bull-spread strategies using pandas, NumPy, and matplotlib.
- **Machine learning:** constructed MA, RSI, MOM, EMA, and MACD features and completed random-forest, GridSearchCV, bagging, boosting, voting, and stacking experiments.

METHODS AND TECHNICAL SKILLS

Integrated assessment	GCAM Global; DICE/RICE family; emissions constraints; technology-support scenarios; carbon-price analysis
Modeling software	GAMS; GCAM XML configuration; model execution; database queries; EViews
Programming	Python; pandas; NumPy; matplotlib; scikit-learn
Quantitative methods	Regression; DEA-BCC; Malmquist index; factor analysis; Tobit models; questionnaire analysis
Research workflow	Literature synthesis; scenario construction; data processing; visualization; policy interpretation; academic writing

RESEARCH INTERESTS

Integrated assessment modeling; climate policy evaluation; industrial decarbonization; carbon pricing; technology-policy interactions; energy-system transition; climate and air-quality co-benefits.